

ADITI GARG

AI Engineer | RAG | LLMs | Agentic AI | Gen AI

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SUMMARY

AI Engineer with experience in Python development and hands-on expertise in building LLM applications, Retrieval-Augmented Generation (RAG) systems, and multi-agent AI frameworks. Proficient in Generative AI, LangChain, LangGraph, CrewAI, prompt engineering, and machine learning. Currently pursuing a Post Graduate Diploma in AI/ML and Agentic AI Engineering at IIT Gandhinagar, focused on building reliable, scalable, and production-ready AI solutions.

TECHNICAL SKILLS

Languages	Python, SQL
AI Frameworks	LangChain, LangGraph, CrewAI, Hugging Face, FastAPI
Generative AI	Large Language Models (LLMs), Prompt Engineering, LoRA Fine-tuning
RAG & Retrieval	Retrieval-Augmented Generation (RAG), Embeddings, Chunking, Hybrid Search, Semantic Search, Query Rewriting, Re-ranking
Machine Learning	PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Feature Engineering, Model Evaluation
Agentic AI	Multi-Agent Systems, Agentic Workflows, Tool Calling, Agent Memory, Model Context Protocol (MCP)
Vector Databases	Qdrant, ChromaDB, FAISS
Developer Tools	Git, Docker, Cursor, Lovable, VS Code

PROFESSIONAL EXPERIENCE

Capgemini <i>Python Developer</i>	Pune, India Mar 2025 – Aug 2025
- Diagnosed, debugged, and refactored Python application logic to resolve 200+ software defects, improving application stability and code quality. - Developed 15+ complex SQL queries and automated data validation workflows, reducing manual review effort by 40% and improving data accuracy. - Performed production support, root cause analysis, and issue resolution for high-priority incidents, maintaining a 98%+ on-time resolution rate.	
Vaco Binary Semantics LLP <i>Associate Analyst</i>	Gurugram, India Nov 2023 – Oct 2024
- Maintained and enhanced 3 business-critical Python applications used by 20+ daily users, improving application performance and reducing downtime by 30%. - Automated SQL-based reporting workflows and data extraction processes, saving over 3 hours of manual effort each week. - Collaborated with cross-functional teams to troubleshoot production issues, implement bug fixes, and support on-going application maintenance.	
Infoxen Technologies <i>Software Developer</i>	Noida, India Jun 2023 – Oct 2023
- Found and fixed errors in Python code to keep the company's applications running smoothly for daily use. - Wrote and checked SQL queries to fetch, update, and clean data, and fixed issues found while testing. - Helped maintain existing applications by updating code, testing changes, and fixing small bugs reported by users.	

TECHNICAL PROJECTS

NCERT Class 9 Science QA Chatbot

GitHub: github.com/divyaitiz/ncert-class9-science-llm

Technologies: *Python, LangChain, ChromaDB, Groq, Llama 3, BGE-large, Cross Encoder, Streamlit*

- Developed an end-to-end Retrieval-Augmented Generation (RAG) chatbot using LangChain, Groq-hosted Llama 3, and ChromaDB vector database to answer questions from 12 NCERT Science chapters.
- Built a complete retrieval pipeline comprising document ingestion, custom PDF layout extraction, tokenization, semantic chunking, Transformer-based BGE-large embeddings, and storage of 700+ document chunks in the ChromaDB vector database for efficient semantic retrieval.
- Implemented a retrieve → re-rank → generate workflow using dense retrieval, cross-encoder re-ranking, query rewriting, prompt templates, retrieval evaluation, and hallucination monitoring, generating grounded responses with source citations.

Customer Intelligence Platform

GitHub: github.com/aditi23garg/customer-intelligence-platform

Technologies: *Python, FastAPI, LangChain, FAISS, Gemini Embeddings, XGBoost, MLflow*

- Developed an AI-powered application combining machine learning and Retrieval-Augmented Generation (RAG) to analyze 5,000+ customer complaint records.
- Designed a semantic retrieval pipeline using Gemini embeddings, FAISS vector indexing, hybrid search, prompt templates, dynamic query rewriting, and semantic retrieval to improve search relevance and response quality.
- Integrated an XGBoost classification model with the RAG workflow through a unified FastAPI service and leveraged MLflow for experiment tracking, model evaluation, and experiment management.

Nexus Supply Co. Operations Assistant

GitHub: github.com/aditi23garg/operations-assistant

Technologies: *Python, CrewAI, FastMCP, Ollama, Qwen 2.5, Model Context Protocol (MCP), Pytest*

- Designed a CrewAI-based multi-agent system using FastMCP to orchestrate Researcher and Writer agents across multiple Model Context Protocol (MCP) tools for automating operational workflows.
- Built an agent workflow involving task decomposition, planning, prompt templates, tool calling, agent memory, and multi-agent collaboration to generate accurate, citation-backed operational reports.
- Improved agent reliability by implementing a structured pre-processing layer for secure tool execution, reducing planning failures and validating architecture stability using 28 unit and integration tests with Pytest.

EDUCATION

IIT Gandhinagar

Post Graduate Diploma in AI/ML & Agentic AI Engineering

Feb 2026 – July 2026

Bharat Institute of Technology

B.Tech in Electronics & Communication Engineering

Aug 2016 – Oct 2020

CERTIFICATIONS

- Anthropic – Introduction to Model Context Protocol (MCP)
- BCG – GenAI Job Simulation (Forage)
- IBM – Data Engineering Foundations Specialization
- Jobaaj – Applied Data & Business Analytics Micro-Degree
- Codechef – Advanced Python, Advanced SQL, GitHub, NumPy, Pandas